

Rheumatic Fever: Approach to a Case of Fever and Palpitation

Dr. Muhammad Reihan, Internal Medicine | Source: Fever & Palpitation – Rheumatic fever (21 slides)

Legend: ★ = high-yield · [added] = not on slides. Images marked "Slide N" come from the lecture; "Added diagram" = drawn for these notes.

🔗 **Bridge from Infective Endocarditis (same lecture block):** Both are "**fever + palpitation + murmur**" problems, but the mechanisms are opposite. **IE** is bacteria growing **on** the valve.

Rheumatic fever is **sterile autoimmune** inflammation weeks after a strep throat. The links:

- Rheumatic valves are "**acquired valvular dysfunction**", the moderate-risk IE group.
- RF scarring is the **#1 cause of mitral stenosis** (next lecture: **Valvular Heart Disease**).
- Pericarditis from the **Pericardial lecture** is one part of rheumatic **pancarditis**.

1. Title / Objectives

Domain	Objective	Likely tested as
Knowledge	Pathophysiological basis and etiology	GAS pharyngitis → molecular mimicry
Knowledge	★ Epidemiology, manifestations, complications	Jones criteria ; mitral valve
Skill	★ Comprehensive management plan	Penicillin V eradication; benzathine penicillin G prophylaxis durations

Lecture flow: Opening case → Definition → Etiology (GAS; ARF vs RHD) → Pathophysiology → Pathology (Aschoff bodies) → Clinical features (constitutional, joints, heart, CNS, skin) → Jones criteria → Confirmation of GAS → Management (eradication) → Prevention (primary / secondary) → Summary.

2. Main Content (Lecture Order)

2.0 Opening Case (Slides 4-5)

- **Patient:** 14-year-old boy with **3 days** of fever, malaise and **severe right knee pain and swelling**.
- **History:** abdominal pain and **epistaxis**. **Left ankle** pain and swelling **5 days ago, now resolved** (= **migratory**). **Sore throat 3 weeks ago** while camping, treated **symptomatically only**.
- **Exam:** T 38.7 °C, HR 119, BP 90/60. Swollen tender right knee. ★ **Painless 3-4 mm nodules over the elbow**. Cardiopulmonary exam normal.
- **Labs:** Hb 12.3, WBC 11 800, ★ **ESR 63**.
- **Knee aspirate:** clear straw-coloured fluid, **WBC 1350 with 17% neutrophils**, Gram stain negative (= non-septic).

→ *Most likely diagnosis?* (A) **acute rheumatic fever** · (B) septic arthritis · (C) ALL · (D) IE · (E) Kawasaki · (F) JIA. (Answered in §7, Q1.)

2.1 Definition & Epidemiology (Slide 6)

Rheumatic fever = an **inflammatory sequela** involving the:

- **heart**
- **joints**
- **skin**
- **CNS**

It occurs ★ **2-4 weeks after an UNTREATED group A streptococcus (GAS) infection.**

- **Prevalence: more common in resource-limited countries** ([added] overcrowding, poverty).
- **Peak age 5-15 years** (slide 11 figure).

2.2 Etiology (Slides 7-8) ★

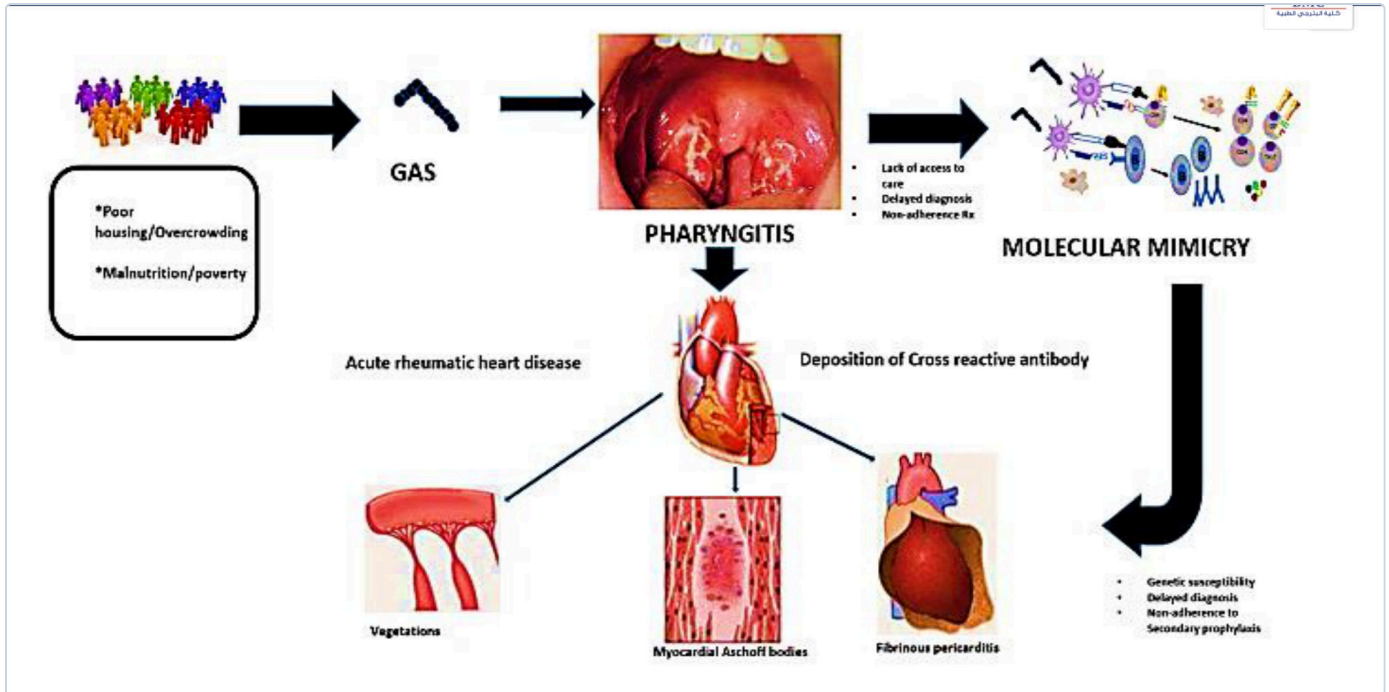
- Previous infection with **group A β -haemolytic streptococcus (GAS) = *Streptococcus pyogenes*.**
- Usually ★ **acute tonsillitis or pharyngitis ("strep throat").**
- ★ **GAS SKIN infections** (erysipelas, impetigo, cellulitis) lead to **post-streptococcal glomerulonephritis**, NOT rheumatic fever.

Term	Meaning
Acute rheumatic fever (ARF)	Multisystem disease from an autoimmune reaction to GAS infection
Rheumatic heart disease (RHD)	Valvular damage that persists after the other ARF features have gone

Mnemonic: Throat → heart; skin → kidney. (PSGN can follow either, but RF follows only pharyngitis.)

2.3 Pathophysiology (Slide 9)

1. **Risk factors:** poor housing/overcrowding, malnutrition/poverty.
2. **GAS pharyngitis.**
3. Lack of access to care, delayed diagnosis, no adherence.
4. ★ **Molecular mimicry:** antibodies against streptococcal antigens ([added] **M protein**) **cross-react** with host tissue ([added] cardiac myosin, valve endothelium, synovium, basal ganglia). **Type II hypersensitivity** (slide 11).
5. **Deposition of cross-reactive antibody** → acute rheumatic heart disease: **vegetations** (valvulitis), **myocardial Aschoff bodies**, **fibrinous pericarditis**.
6. Genetic susceptibility, recurrent GAS, no secondary prophylaxis → **chronic RHD**.

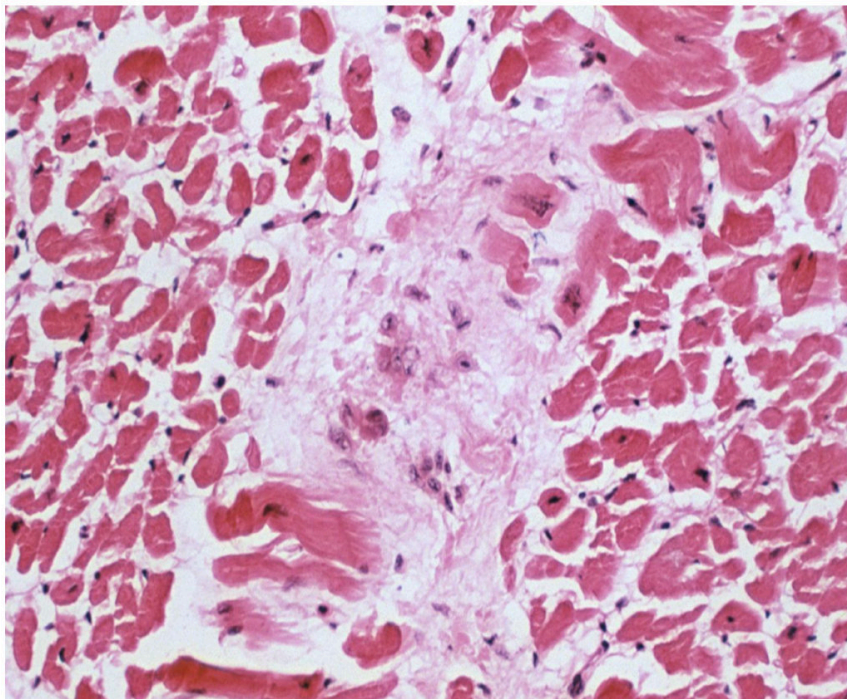


Slide 9: pathophysiology (GAS pharyngitis → molecular mimicry → carditis)

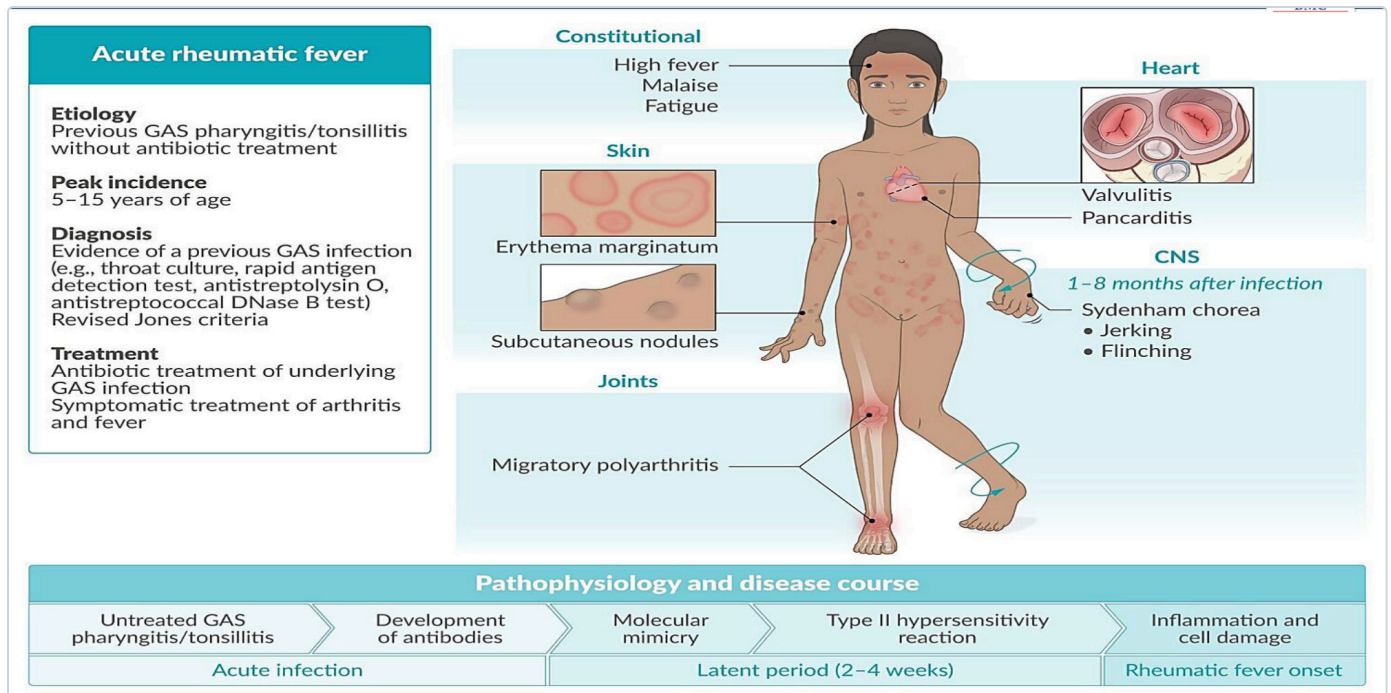
2.4 Pathology (Slide 10)

★ **Aschoff bodies** (myocardium) = **fibrinoid necrosis with histiocytic infiltrate**.

[added] They contain **Anitschkow cells** ("caterpillar cells": macrophages with wavy, ribbon-like chromatin). Pathognomonic of rheumatic carditis.

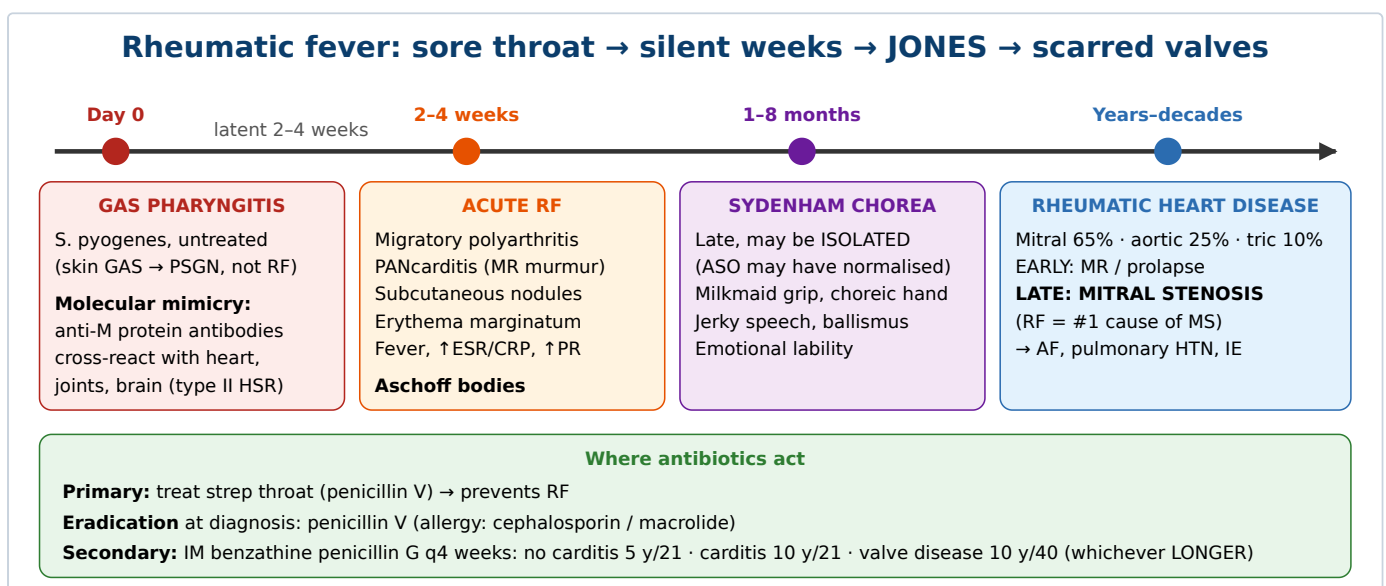


Slide 10: Aschoff body



Slide 11: acute rheumatic fever overview (AMBOSS)

Slide 11 summary: etiology = previous GAS pharyngitis **without antibiotic treatment**; **peak 5-15 years**; diagnosis = evidence of GAS + **revised Jones criteria**; treatment = antibiotics for GAS + **symptomatic treatment of arthritis and fever**. Course: untreated pharyngitis → antibody formation → molecular mimicry → type II hypersensitivity → inflammation, during a **latent period of 2-4 weeks**.



Added diagram: RF timeline from strep throat to rheumatic heart disease

Recap: Cause. ARF = autoimmune sequela 2–4 weeks after **untreated GAS pharyngitis** (skin GAS → PSGN instead). Mechanism: molecular mimicry (type II HSR). Hallmark: Aschoff bodies. ARF is acute and multisystem; RHD is the chronic valve scarring left behind.

Link: Antibodies cross-react with joints, heart, skin and brain, and those targets are the clinical features.

Up next: Clinical features and Jones criteria.

2.5 Clinical Features (Slides 12-14) ★

System	Features
Constitutional	Fever, malaise, fatigue
Joints	★ Migratory polyarthrititis ([added] large joints, hugely responsive to aspirin, no permanent damage)
Heart	★ PANcarditis (endocarditis + myocarditis + pericarditis)
CNS	★ Sydenham chorea
Skin	★ Subcutaneous nodules · ★ erythema marginatum

Heart: valvular lesions mostly on **high-pressure (left) valves**:

Valve	Frequency	Lesion
★ Mitral	~65%	EARLY: mitral regurgitation or prolapse · LATE: mitral STENOSIS. ★ RF is the most frequent cause of mitral stenosis
Aortic	~25%	([added] AR > AS; usually with mitral disease)
Tricuspid	~10%	

CNS: Sydenham chorea (slide 13)

- **Involuntary, irregular, non-repetitive** movements of the limbs, neck, head and/or face.
- ★ Appears **1-8 months after** the inciting infection (late; [added] may be the **only** manifestation, and ASO may be normal by then).
- **Additional motor symptoms:** - **Ballismus** - **Muscle weakness** - **Speech disorders** (slurred / "jerky") - ★ **Milkmaid grip** (rhythmic squeezing) - **Choreic hand**



Slide 13: Sydenham chorea

Skin (slide 14):

- **Subcutaneous nodules:** *[added]* painless, firm, over **extensor surfaces / bony prominences** (elbows, knees, occiput). Associated with **carditis**.
- ★ **Erythema marginatum:** **centrifugally expanding** pink/light-red rash with a **well-defined outer border and central clearing**. **Painless and non-pruritic** (*[added]* trunk and proximal limbs, never the face).



Slide 14: subcutaneous nodules and erythema marginatum

2.6 Diagnostic Criteria: Jones (Slide 15) ★

Mnemonic: "JONES CAFE PAL"

Major (JONES)	Minor (CAFE PAL)
J oint involvement (migratory polyarthritis)	C RP increased
O looks like a heart ♥ = (pan) carditis	A rthralgia
N odules, subcutaneous	F ever
E rythema marginatum	E levated ESR
S ydenham chorea	P rolonged PR interval
	A namnesis (history) of rheumatism (previous RF/RHD)
	L eukocytosis

★ **Diagnosis = evidence of preceding GAS infection** (throat culture growing GABHS **or** elevated ASO titre) + **2 major** criteria, **OR 1 major + 2 minor**.

Mnemonic: "JONES CAFE PAL"

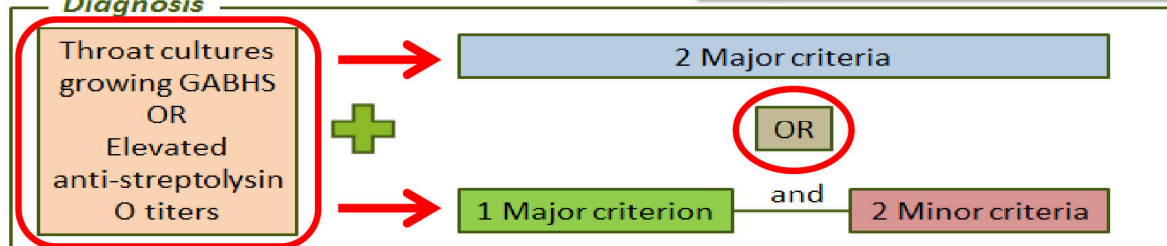
Major Criteria

J	Joint Involvement
O	O looks like a heart = myocarditis
N	Nodules, subcutaneous
E	Erythema marginatum
S	Sydenham chorea

Minor Criteria

C	CRP Increased
A	Arthralgia
F	Fever
E	Elevated ESR
P	Prolonged PR Interval
A	Anamnesis of Rheumatism
L	Leukocytosis

Diagnosis



Slide 15: Jones criteria (JONES CAFE PAL)

[added] Key 2015 AHA revision points:

- Separate **low-risk** and **moderate/high-risk** populations.
- In high-risk populations, **monoarthritis** or **polyarthralgia** can count as major.
- **Subclinical carditis on echo** counts as carditis.
- **Arthralgia cannot be a minor criterion if arthritis is already counted as major.**
- **Prolonged PR cannot be counted if carditis is a major criterion.**
- **Recurrent RF:** 2 major, or 1 major + 2 minor, or **3 minor**.
- **Chorea alone** is enough (GAS evidence not required).

2.7 Confirmation of GAS Infection (Slide 16)

Any of the following:

- ↑ **Antistreptolysin O (ASO) titre**
- ↑ **Anti-streptococcal DNase B (ADB) titre**
- (Slide 15:) **throat culture** growing GABHS
- [added] rapid antigen test

[added] Rising titres on paired samples are best. ADB stays raised longer, which helps for late chorea.

Recap: Diagnosis.

- **Major (JONES):** migratory polyarthritis, pancarditis (mitral early MR → late MS), nodules, erythema marginatum, Sydenham chorea (1–8 months later).
- **Minor (CAFE PAL):** CRP, arthralgia, fever, ESR, prolonged PR, previous RF, leukocytosis.
- **Rule:** GAS evidence (culture/ASO/ADB) + 2 major, or 1 major + 2 minor.

Link: Once diagnosed, treatment has two jobs: eradicate any remaining GAS now, and stop the next attack.

Up next: Management and prevention.

2.8 Management (Slide 17)

GAS eradication:

- ★ **First line: penicillin V.**
- Penicillin allergy:** - **Cephalosporins** for hypersensitivity **without anaphylaxis** - **Macrolides** for severe β -lactam hypersensitivity

[added] Not on the slides but standard:

- Arthritis/fever:** high-dose **aspirin or NSAIDs** (dramatic response).
- Moderate-severe carditis/heart failure:** **corticosteroids** $\pm$ diuretics, ACEI.
- Chorea:** haloperidol / valproate, plus rest.
- Bed rest** while carditis is active.
- Echo** for every patient.

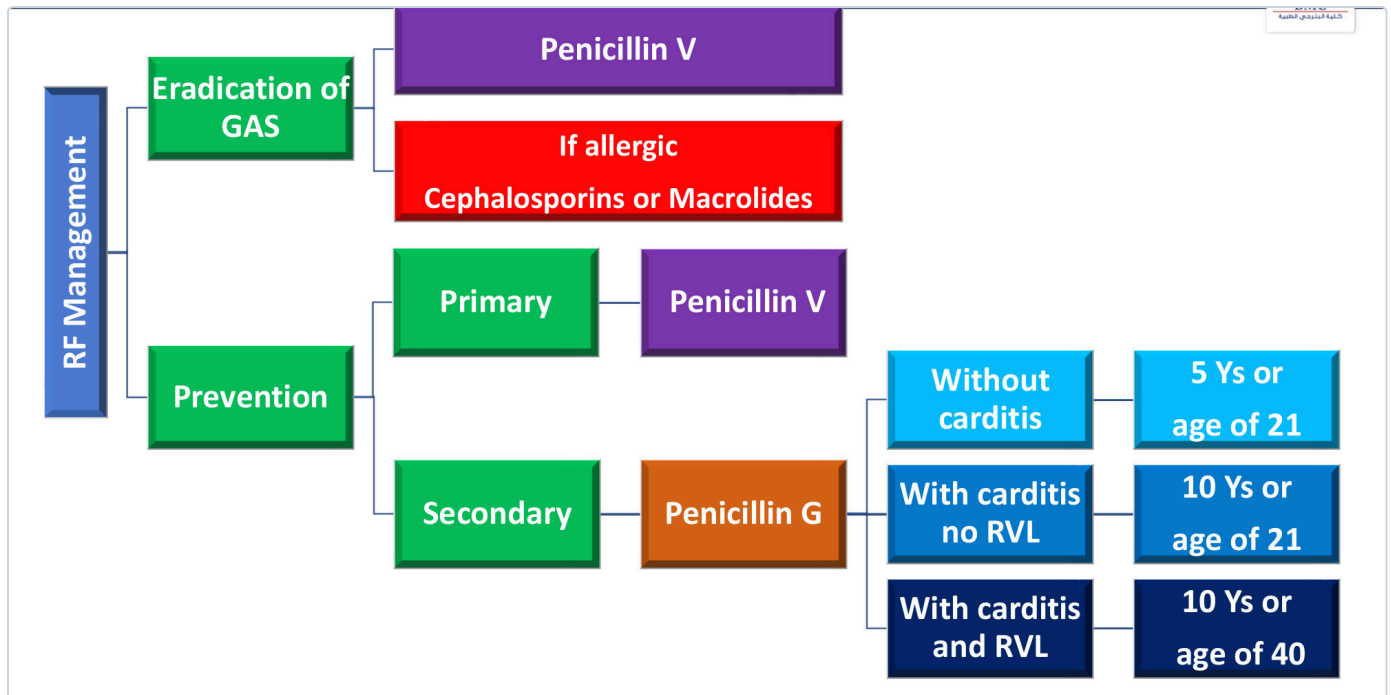
2.9 Prevention (Slides 18-19) ★

Level	What
Primary prevention	Prompt antibiotic treatment of GAS tonsillopharyngitis (e.g., penicillin V) $\rightarrow$ <i>[added]</i> prevents RF if started within 9 days of symptoms
Secondary prevention	★ ALL patients need antibiotic prophylaxis. Drug of choice: IM benzathine penicillin G every 4 weeks

Secondary prophylaxis duration: use whichever gives the **LONGEST** course.

Category	Duration
RF without carditis	5 years or until age 21
RF with carditis, no residual heart disease	10 years or until age 21
RF with carditis AND permanent valvular disease	10 years or until age 40 (<i>[added]</i> sometimes lifelong)

Mnemonic: "5-21, 10-21, 10-40": no heart $\rightarrow$ 5/21; heart healed $\rightarrow$ 10/21; heart scarred $\rightarrow$ 10/40.



Slide 19: RF management summary

Recap: Management. Eradicate GAS with penicillin V (cephalosporin if non-anaphylactic allergy; macrolide if severe allergy). Primary prevention = treat every strep throat. Secondary prevention = IM benzathine penicillin G every 4 weeks for 5 y/21 (no carditis), 10 y/21 (carditis healed) or 10 y/40 (valve disease), whichever is longer. *[added]* Aspirin/NSAIDs for arthritis, steroids for severe carditis.

3. High-Yield Comparison Tables

Table A: Major Criteria in Detail

Major	Frequency <i>[added]</i>	Key detail	Leaves damage?
Polyarthritis	~75% (commonest)	Migratory , large joints, aspirin-responsive	No
Carditis	~50–60%	Pancarditis ; mitral (MR → MS)	Yes (RHD)
Sydenham chorea	~10–30%	1-8 months late; milkmaid grip	No (resolves)
Subcutaneous nodules	Rare	Painless, extensor surfaces; with carditis	No
Erythema marginatum	Rare	Central clearing, non-pruritic, trunk	No

"RF licks the joints but bites the heart." *[added]*

Table B: Rheumatic Fever vs PSGN (same bug, different site)

	Rheumatic fever	PSGN <i>[added detail]</i>
Preceding infection	Pharyngitis only	Pharyngitis or skin (impetigo)
Latency	2-4 weeks	1–2 weeks (throat) / 3–6 weeks (skin)

	Rheumatic fever	PSGN <i>[added detail]</i>
Mechanism	Type II (molecular mimicry)	Type III (immune complex)
Prevented by antibiotics?	Yes	No
Target	Heart, joints, skin, brain	Kidney

Table C: RF Arthritis vs Septic Arthritis vs JIA (opening case)

	ARF	Septic arthritis	JIA
Pattern	Migratory large joints	Single, fixed	Persistent ≥ 6 weeks
Synovial WBC	Inflammatory but modest (1000s), not neutrophil-dominant	> 50 000, > 90% neutrophils , organisms	Inflammatory
Clue	Sore throat 2–4 weeks earlier, nodules, $\uparrow$ ESR	Fever, toxic, Gram stain positive	Morning stiffness, uveitis

Table D: Secondary Prophylaxis

Carditis?	Residual valve disease?	Duration
No	—	5 y or age 21
Yes	No	10 y or age 21
Yes	Yes	10 y or age 40

4. Rules, Exceptions, Patterns

Rule	Exception / nuance
GAS $\rightarrow$ RF	Skin GAS $\rightarrow$ PSGN, not RF
Jones needs GAS evidence	<i>[added]</i> Sydenham chorea alone is diagnostic (ASO may be normal by then)
Arthritis heals	Carditis leaves RHD , the only permanent damage
Mitral lesion = MS	Early = MR; late = MS
Aspirin for arthritis	Arthralgia and arthritis can't both be counted <i>[added]</i>
PR prolongation is a minor criterion	Not counted if carditis is already counted as major <i>[added]</i>
Penicillin V for eradication	Benzathine penicillin G IM monthly for secondary prevention
Prophylaxis stops at 21	With valve disease, continue to 40 (whichever is longer)
Allergy $\rightarrow$ macrolide	Non-anaphylactic allergy $\rightarrow$ cephalosporin first

5. Clinical Correlations

- Child with an untreated sore throat, then migratory knee and ankle arthritis, nodules and high ESR $\rightarrow$ ARF (opening case).**
- Girl with clumsy writing, emotional lability, milkmaid grip; strep throat 4 months ago $\rightarrow$ Sydenham chorea.** Isolated chorea is diagnostic.

- **Young adult from a low-income country with dyspnoea, AF, malar flush, opening snap** → **rheumatic mitral stenosis** (next lecture).
- **New murmur + fever in a patient with RHD** → think **IE** on a rheumatic valve. Rheumatic valves are a moderate-risk IE group.
- **Impetigo followed 3 weeks later by cola-coloured urine and oedema** → **PSGN**, not RF.
- **Prolonged PR on ECG during the acute illness** → minor criterion. *[added]* It does not by itself mean carditis.

6. Rapid Revision (Last-Day)

1. RF = inflammatory sequela (heart, joints, skin, CNS) **2-4 weeks after untreated GAS pharyngitis**; resource-limited countries; peak **5-15 years**.
2. **Skin GAS** → **PSGN, not RF**.
3. **ARF** = acute multisystem autoimmune disease. **RHD** = persistent valve damage.
4. **Molecular mimicry** (type II HSR). **Aschoff bodies** = fibrinoid necrosis + histiocytes.
5. **Migratory polyarthritides · pancarditis · Sydenham chorea · subcutaneous nodules · erythema marginatum.**
6. Valves: **mitral 65%** (early MR/prolapse → late **MS**), **aortic 25%**, **tricuspid 10%**. **RF = #1 cause of MS.**
7. Chorea: involuntary, irregular, non-repetitive; **1-8 months later**; ballismus, weakness, jerky speech, **milkmaid grip**, choreic hand.
8. Erythema marginatum: expanding, sharp outer edge, central clearing, **painless, non-pruritic**.
9. **JONES CAFE PAL. GAS evidence + 2 major, or 1 major + 2 minor.**
10. GAS evidence: **throat culture, ↑ ASO, ↑ anti-DNase B.**
11. Eradication: **penicillin V**; allergy → **cephalosporin** (non-anaphylactic) or **macrolide** (severe).
12. Primary prevention: treat strep throat (penicillin V).
13. Secondary: **IM benzathine penicillin G every 4 weeks: 5 y/21 · 10 y/21 · 10 y/40**, whichever is longest.

7. Exam-Style Questions

Q1 (lecture case). A 14-year-old with a sore throat 3 weeks ago (symptomatic treatment only) has migratory arthritis (left ankle, then right knee), fever, painless elbow nodules and ESR 63. Synovial fluid: WBC 1350, 17% neutrophils, Gram stain negative. What is the most likely diagnosis? A. **Acute rheumatic fever** · B. Septic arthritis · C. ALL · D. Infective endocarditis · E. Kawasaki disease · F. JIA

Answer: A. *[reasoning added; key not on slides]*

- **Two major criteria** (migratory polyarthritides + subcutaneous nodules) plus minors (fever, ↑ ESR, leukocytosis) after an untreated pharyngitis.
- GAS confirmation (ASO/culture) would complete the Jones criteria.
- Wrong options:
 - **Septic arthritis:** the fluid is non-septic (low WBC, few neutrophils, no organisms) and the joints migrate.
 - **ALL:** epistaxis and bone pain could fit, but the CBC is normal.
 - **Kawasaki:** no conjunctivitis, mucositis or rash, and he is too old.
 - **JIA:** needs ≥ 6 weeks.
 - **IE:** no murmur and no risk factor.

Q2. Which streptococcal infection is LEAST likely to be followed by rheumatic fever? A. Tonsillitis · B. Pharyngitis · C. **Impetigo** · D. Scarlet fever · E. Untreated strep throat

Answer: C.

- **Skin GAS infections lead to PSGN rather than RF** (slide 7).

Q3. A 12-year-old had ARF with carditis but **no residual valve disease** on echo. How long should secondary prophylaxis last? A. 5 years or to age 21 · B. **10 years or to age 21, whichever is longer** · C. 10 years or to age 40 · D. Lifelong · E. Until ASO normalises

Answer: B (slide 18).

- Age 21 is 9 years away; 10 years takes him to age 22. The longer option wins, so he continues **to age 22**.

Q4. What is the drug of choice for secondary prevention of RF? A. Oral penicillin V daily · B. **IM benzathine penicillin G every 4 weeks** · C. Oral azithromycin weekly · D. Amoxicillin before dental work · E. IV ceftriaxone monthly

Answer: B (slide 18).

- **D** is IE prophylaxis, a different indication.

Q5. A 35-year-old woman who had RF as a child now has exertional dyspnoea, haemoptysis, a loud S1, an opening snap and a mid-diastolic apical rumble. Which valve lesion does she have, and why? A. MR, early rheumatic lesion · B. **MS, late rheumatic lesion** · C. AS, calcific · D. AR, aortic root dilatation · E. TR, functional

Answer: B.

- Rheumatic **mitral** disease starts as **MR** and ends as **MS** (slide 12). RF is the most common cause of MS.

8. Exam Pitfalls & Traps

- **RF follows pharyngitis only.** Skin infection → PSGN.
- **The minor criteria on slide 15 include "Anamnesis of rheumatism" and "Leukocytosis".** These come from the older Jones version. The **2015 AHA** minors are fever, arthralgia (polyarthralgia in high-risk populations), **ESR ≥ 60 / CRP ≥ 3 mg/dL**, prolonged PR. For the local exam, learn the lecturer's **CAFE PAL** list.
- **Double counting:** arthritis (major) and arthralgia (minor) can't both be counted; carditis (major) and prolonged PR (minor) can't both be counted *[added]*.
- **Chorea can appear alone** and late, when ASO may have normalised. Don't reject the diagnosis.
- **Early mitral lesion = MR**, not MS. MS is the **late** scar.
- **Prophylaxis durations:** "whichever is LONGER". A child of 8 without carditis → 5 years would end at 13, but "until 21" is longer, so prophylaxis runs to 21.
- **Penicillin V (oral)** eradicates and gives primary prevention. **Benzathine penicillin G (IM)** is for secondary prevention. Don't swap them.
- **Allergy:** cephalosporin for non-anaphylactic allergy; macrolide for **severe** β -lactam hypersensitivity.

- **Management slides omit anti-inflammatory therapy.** Know aspirin/NSAIDs for arthritis and steroids for severe carditis *[added]*.
- **Slide 15 shows a heart symbol** in the "O" of JONES. It stands for **carditis**: "O looks like a heart".